INTERNSHIP TASK SUBMISSION

Data Analysis of Tamil Nadu Public Transport Services

For Improving Accessibility and Operational Planning

Week 1 Task — Data Analysis Planning and Strategy

Submitted by

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B.Tech — Artificial Intelligence and Data Science | Batch 2026

May 2026

1. Introduction

Public transport is a critical infrastructure component supporting daily mobility across Tamil Nadu's urban and rural regions. A significant proportion of the state's population depends on the State Express Transport Corporation (SETC) bus network for inter-district and intra-district travel. As digital governance systems expand, public transport operations generate increasing volumes of structured data encompassing routes, service timings, frequency metrics, and depot-level operational information.

Strategic analysis of these publicly available datasets enables evidence-based identification of service distribution patterns, accessibility disparities, and operational inefficiencies. This document presents a comprehensive data analysis planning framework, designed to systematically collect, preprocess, analyze, and interpret Tamil Nadu transport data — with the objective of generating actionable insights for digital governance and public service planning.

2. Problem Statement

> **Core Problem**
> Tamil Nadu's public transport network spans hundreds of districts, routes, and service frequencies. Disparities in service concentration and route distribution may result in differential accessibility across regions — with certain districts experiencing underservice while others demonstrate high transport density. The absence of structured analytical frameworks limits the capacity of planners to identify these gaps using available public data.

This project addresses the need to analyze SETC transport datasets to surface service distribution inequities, identify high-frequency corridors, and understand connectivity patterns across major transport hubs — enabling data-driven intervention strategies within the e-governance framework.

3. Objectives of the Study

The following analytical objectives define the scope and direction of this data analysis initiative:

- Quantify variation in service frequency and route distribution across districts.
- Identify districts exhibiting lower transport service availability relative to the state average.
- Identify districts with disproportionately higher service concentration.
- Determine routes with the highest operational frequency across the network.
- Analyze frequency and connectivity of routes linking major transport hub districts.
- Examine timing distribution patterns between primary transport hubs to assess scheduling efficiency.

4. Analytical Hypotheses

The following testable hypotheses are formulated to guide the analysis and structure the interpretation of results:

- **H1:** Districts with a lower count of registered routes and service timings will demonstrate measurably reduced transport service availability.
- **H2:** Districts classified as major transport hubs will exhibit significantly higher bus service frequency relative to non-hub districts.
- **H3:** Major transport hub districts will demonstrate stronger inter-district route connectivity compared to peripheral districts.

5. Public Data Sources

This analysis is grounded in publicly accessible government transport datasets. The primary dataset identified for this project is the SETC Bus Timings Dataset.

SETC Bus Timings Dataset

The SETC Bus Timings dataset provides granular operational information across the Tamil Nadu public transport network. Key data fields include:

- Depot Name — operational depot responsible for the route
- Route Number — unique identifier for each bus route
- Source and Destination — origin and terminus locations

- Route Length — total distance covered by the service
- Service Type — classification of service (express, ordinary, etc.)
- Number of Services — frequency count for each route
- Departure Timings — scheduled departure times per route

Platform	Open Government Data (OGD) Platform India
Domain	Open Government Data (OGD) Platform Tamil Nadu
URL	data.gov.in/catalog/setc-bus-routes-services-and-timings
License	National Data Sharing and Accessibility Policy (NDSAP)
Published On	24 January 2018
Last Updated	19 July 2019
Format	CSV (Comma-Separated Values)

6. Data Collection Process

The data collection workflow is structured to ensure systematic acquisition and initial organization of transport datasets prior to analysis:

- **Step 1:** Access and download SETC transport datasets from Tamil Nadu government and open data portals in CSV format.
- **Step 2:** Extract transport-relevant fields — routes, timing schedules, service frequency, and depot metadata.
- **Step 3:** Organize extracted data into a structured tabular format with clearly labeled columns and consistent data types.
- **Step 4:** Conduct an initial data inventory to document field names, record counts, and data coverage across districts.
- **Step 5:** Prepare the structured dataset for entry into the preprocessing pipeline.

7. Data Preprocessing Plan

Prior to analysis, all collected datasets will undergo a rigorous preprocessing workflow to ensure data integrity and analytical reliability:

- Conduct an initial quality audit — assess completeness, consistency, and structural integrity of the raw dataset.
- Identify and address missing values — apply appropriate imputation strategies or flag records for exclusion.
- Detect and eliminate duplicate records — ensure each route-timing combination is represented once.
- Standardize formatting — normalize route names, district labels, and timing values to a uniform schema.
- Validate and reconcile data types — ensure numerical fields (route length, service count) and categorical fields (depot names, service types) are correctly formatted.
- Produce a cleaned, analysis-ready dataset with documented preprocessing decisions.

8. Analysis Techniques and Methodology

The analytical framework incorporates four complementary methodological approaches to extract comprehensive insights from the transport dataset:

8.1 Descriptive Analysis

Establishes the baseline profile of the transport network by computing summary statistics — route counts, average service frequencies, and district-wise service distribution. This phase provides a factual foundation for all subsequent comparative and inferential analysis.

8.2 Comparative Analysis

Evaluates relative performance across districts by benchmarking service frequency, route density, and transport availability metrics. Districts are ranked and segmented to identify high-performing and underserved regions within the network.

8.3 Frequency Analysis

Examines temporal and operational patterns in route scheduling — identifying high-frequency corridors, peak operational windows, and the scheduling density between major transport hubs. This analysis supports hypotheses regarding hub connectivity and service concentration.

8.4 Visualization-Based Analysis

Translates quantitative findings into interpretable visual formats — bar charts, distribution plots, heatmaps, and route frequency graphs — to facilitate stakeholder comprehension and support reporting deliverables.

9. Tools and Technologies

The following tools have been selected based on their suitability for transport data analysis and alignment with open-source analytical workflows:

Tool / Technology	Role in Analysis
Microsoft Excel	Initial dataset inspection, filtering, and manual data quality review
Python 3.x	Primary language for data processing, transformation, and analytical operations
Pandas	Structured handling and manipulation of CSV-format transport datasets
Matplotlib	Generation of graphs, bar charts, and frequency distribution visualizations
Power BI	Optional dashboard creation for interactive transport data representation

10. Expected Outcomes

Upon completion of the analysis, the following deliverables and insights are anticipated:

- Ranked identification of districts with below-average transport service availability.
- Mapped identification of districts with disproportionately high service concentration.
- A curated list of high-frequency routes across the Tamil Nadu transport network.
- A connectivity profile of major transport hub districts and their inter-district linkages.
- Visualized route and timing distribution patterns across the network.
- A data-backed foundation for operational planning recommendations within the e-governance framework.

- An analytical report supporting evidence-based decision-making for transport governance authorities.

11. Assumptions and Anticipated Challenges

11.1 Working Assumptions

- The SETC dataset accurately reflects current operational routes and scheduled service timings.
- Departure timings in the dataset correspond to actual, active service schedules rather than historical records.
- Available public datasets provide sufficient district-level coverage to support meaningful comparative analysis.

11.2 Anticipated Challenges

- Incomplete or missing data fields — particularly for rural or lower-frequency routes.
- Dataset currency risk — publicly available data may not reflect the most recent route additions or service modifications.
- Limited granularity in district-level coverage — certain administrative subdivisions may be underrepresented.
- Absence of passenger ridership data — limiting the ability to correlate service supply with actual demand.
- Formatting inconsistencies — variations in naming conventions across depot, route, and district fields requiring manual standardization.

12. Project Timeline

The following phased timeline governs the delivery of all analytical work over the internship period:

Phase	Key Activities
Week 1	Dataset identification, acquisition, and initial exploration of SETC transport data

Phase	Key Activities
Week 2	Data preprocessing — handling missing values, standardizing formats, and structuring datasets
Week 3	Analytical processing — descriptive, comparative, and frequency analysis with visualizations
Week 4	Insight synthesis, report preparation, and final deliverable submission

13. Conclusion

Public transport infrastructure serves as a foundational element of daily mobility and economic participation across Tamil Nadu. Systematic analysis of publicly available SETC transport datasets presents a structured opportunity to surface service distribution inequities, evaluate operational patterns, and strengthen the evidence base for digital governance decision-making.

This strategic plan establishes a clear analytical framework — encompassing data acquisition, preprocessing, multi-method analysis, and visualization — to deliver actionable insights within the defined internship timeline. The outcomes of this analysis are designed to contribute meaningfully to transport accessibility evaluation and operational planning within the broader context of e-governance in Tamil Nadu.